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STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION AND FIFTH REVIEW UNDER THE POLICY COORDINATION INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS[1]

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The Debt Sustainability Analysis (DSA) was prepared jointly by the staffs of the International Monetary Fund and the International Development Association.

Rwanda: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate* [2]
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited space to absorb shocks*
Application of judgment	*No*

The updates to the Bank/Fund assessment of Rwanda's debt sustainability analysis indicate a moderate risk of external and overall public debt distress. The current debt-carrying capacity is consistent with a classification of 'strong'. The baseline scenario is based on the macroeconomic projections under the 5th PCI review, assuming a stronger growth rebound and larger fiscal deficit than was expected in the December 2020 DSA, followed by an ambitious post-pandemic medium-term fiscal consolidation path underpinned by policies under the PCI. Compared to the 4th PCI review (July 2021 update), the baseline incorporates the one-off use of a significant portion of the recent SDR allocation in FY2021/22 and Eurobond issuance in August 2021 to buyback an 85 percent of the 2013 Eurobonds with bullet payment in 2023, which help to smooth out the debt service profile. The debt indicators and stress tests suggest that while liquidity risks have been moderated, solvency risks remain elevated, with limited buffers to cushion external and growth shocks. The country remains susceptible to adverse market conditions, with the risk mitigated by adequate reserves and access to external official financing at concessional terms. The authorities are encouraged to further strengthen their debt management capacity to mitigate heightened risks in the context of the uncertainties surrounding the COVID-19 crisis and implement the fiscal consolidation strategy as envisaged under the PCI, supported by credible measures to bring debt to lower levels with sufficient buffers to absorb shocks. Strengthening the identification, assessment, and management of fiscal risks, also one of the pillars under the PCI, is essential to mitigate debt surprises.

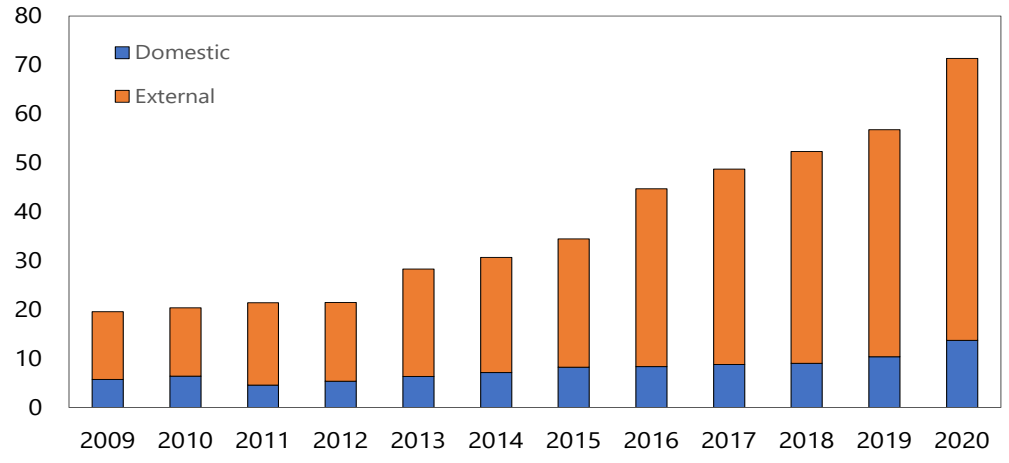
[1] This debt sustainability analysis was conducted using the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (LIC-DSF) that was approved in 2017. The fiscal year for Rwanda is from July to June; however, this DSA is prepared on a calendar year basis.

[2] Rwanda's debt-carrying capacity remains strong as its Composite Indicator is 3.19, which is based on the 2021 October WEO and the 2020 CPIA that was released in July 2021, remains above the upper threshold value of 3.05. Debt coverage includes the central government and state-owned enterprises (SOEs). There has been no change in debt coverage since the December 2020 DSA.

BACKGROUND

1. Rwanda's public and publicly-guaranteed (PPG) external debt-to-GDP ratio increased by 33 percentage points of GDP over the last 6 years, driven by borrowing to meet the development needs envisaged in the National Strategy for Transformation (NST), but also from the robust COVID-19 response. The development needs are supported by a long-planned comprehensive public investment strategy, including three large projects to support trade and tourism through a series of public-private partnerships (PPPs) and external guarantees outside the budgetary central government. The three large projects include the construction of the Kigali Convention Center (KCC), completed in 2016, the expansion of the national airline, RwandAir, that is now completed, and the construction of a new airport in the Bugesera district of eastern Rwanda. These developments contributed to PPG external debt to increase by 21.7 percentage point in the five years preceding the COVID crisis. At the same time, the increase in the fiscal deficit due to revenue shortfalls and a scaling up in spending to address the COVID-19 crisis led to sharp debt increase in 2020 by an additional 11.3 percentage point. As a result, external PPG debt has risen from 22.6 percent of GDP in 2014 to 55.6 percent in 2020 (Text Figure 1). It continues to be dominated by multilateral lending on concessional terms (Text Table 1, Text Figure 2), resulting in a present value (PV) of external PPG debt-to-GDP ratio of 33.6 percent in 2020. Total PPG debt stood at 71.3 percent of GDP in 2020, which is higher than the 2020 DSA projections (estimated at 65.9 percent of GDP) due to lower-than-expected GDP growth and a higher fiscal deficit stemming from pandemic-related spending. In August 2021, Rwanda took advantage of the favorable market conditions and issued US$620 million Eurobond at 5.5 percent yield, partly to buyback the 2013 Eurobonds maturing in 2023. Interest rates on domestic T-bills and T-bonds range from 6.7 percent (28 days) to 12.5 percent (15 years).

Text Figure 1. Total PPG Debt
(Percent of GDP)



Sources: Rwandan authorities and IMF Staff Calculations

Text Table 1. Rwanda: Decomposition of Public Debt and Debt Service by Creditor, 2020–22[1]

	Debt Stock (end of period)			Debt Service					
	2020			2020	2021	2022	2020	2021	2022
	(In US$ mn)	(Percent total debt)	(Percent GDP)	(In US$ mn)			(Percent GDP)		
Total	7,149.8	100.0	71.3	498.2	1,141.2	432.0	5.1	10.4	3.5
External	5,571.8	77.9	55.6	219.3	722.4	181.8	2.1	6.5	1.5
Multilateral creditors[2]	4,190.5	58.6	41.8	62.4	95.8	117.3	0.6	0.9	1.0
IMF	415.3	5.8	4.1	22.3	37.6	46.7	0.2	0.3	0.4
World Bank	2,437.1	34.1	24.3	21.4	37.0	47.4	0.2	0.3	0.4
ADB/AfDB/IADB	1,017.2	14.2	10.1	11.2	13.5	15.5	0.1	0.1	0.1
Other Multilaterals	320.9	4.5	3.2	7.4	7.7	7.6	0.1	0.1	0.1
o/w: IFAD	162.0	2.3	1.6						
BADEA	76.0	1.1	0.8						
Bilateral Creditors	640.0	9.0	6.4	35.6	38.1	37.8	0.3	0.3	0.3
Paris Club	178.1	2.5	1.8	0.0	0.0	0.0	0.0	0.0	0.0
o/w: JICA	113.1	1.6	1.1						
AFD	49.2	0.7	0.5						
Non-Paris Club	461.9	6.5	4.6	35.6	38.1	37.8	0.3	0.3	0.3
o/w: EXIM-CHINA	258.0	3.6	2.6						
SFD	74.0	1.0	0.7						
Bonds	428.6	6.0	4.3	26.5	363.0	4.0	0.3	3.3	0.0
Commercial creditors[3]	312.7	4.4	3.1	94.8	225.6	22.7	0.9	2.0	0.2
Other international creditors	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Domestic	1,578.0	22.1	15.7	278.9	418.8	250.2	2.7	3.8	2.1
Held by residents, total	1,578.0	22.1	15.7	278.9	418.8	250.2	2.7	3.8	2.1
Held by non-residents, total	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
T-Bills	461.4	6.5	4.6	0.0	0.0	0.0	0.0	0.0	0.0
Bonds	708.8	9.9	7.1	256.1	388.0	214.0	2.5	3.5	1.8
Loans	407.8	5.7	4.1	22.8	30.8	36.2	0.2	0.3	0.3
Memo items:									
Collateralized debt[4]	N/A	N/A	N/A						
Contingent liabilities[5]	N/A	N/A	N/A						
Nominal GDP (US$ million)				10,022	10,831	11,475			

1/As reported by country authorities according to their classification of creditors, including by official and commercial. Debt coverage is the same as the DSA.
2/Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears).
3/ A breakdown by creditor is currently not available. Information will be provided in the next DSA.
4/Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt. Collateralization entails a borrower granting liens over specific existing assets or future receivables to a lender as security against repayment of the loan. Collateral is "unrelated" when it has no relationship to a project financed by the loan. An example would be borrowing to finance the budget deficit, collateralized by oil revenue receipts. See the joint IMF-World Bank note for the G20 "Collateralized Transactions: Key Considerations for Public Lenders and Borrowers" for a discussion of issues raised by collateral.
5/Guaranteed debt is included within public sector perimeter for DSA purposes.

Sources: Rwandan authorities and IMF Staff Calculations

2. The DSA covers the central government, guarantees, and state-owned enterprises (Text Table 2). The Ministry of Finance and Economic Planning (MINECOFIN) publishes annual debt data, covering domestic and external debt of the central government, broken down by multilateral, bilateral and commercial debt, as well as information on both domestic and external guarantees and external debt held by all state-owned enterprises (SOEs). There is no debt stemming from extra budgetary funds, long term central bank financing of the government, nor the state-owned social security fund. The local government debt is not covered but the existing stock to date is marginal, and its contracting is subject to approval by MINECOFIN. External debt is defined on a currency basis. The contingent liabilities shock (7 percent of GDP) accounts for the realization of liabilities from corporations where the government has a minority stake and the possible incidence of a financial crisis.

Text Table 2. Rwanda: Coverage of Public and Publicly-Guaranteed Debt and Parameters for Contingent Liability Shocks for the Tailored Stress Test

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

1	The country's coverage of public debt	The central government plus social security and extra budgetary funds, central bank,		
		Default	Used for the analysis	Reasons for deviations from the default
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2	
4	PPP 2/	35 percent of PPP stock	0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)	7.0		

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

2/ When PPP stock is less than 3 percent of GDP, as reflected in the World Bank's database, then test is set to zero. Rwanda's PPP stock is shown as 2.6 percent of GDP.

UNDERLYING ASSUMPTIONS

3. The macroeconomic assumptions underlying the baseline scenario reflect recent economic developments and policies, including the impact of the COVID-19 pandemic, consistent with the fifth PCI review macroeconomic framework. The major differences between the current assumptions and those underlying the last full DSA in 2020 in the context of the Third Review of the PCI are as follows: (i) a larger economic contraction in 2020 followed by a stronger growth rebound in 2021, reflecting resilience in agricultural production and a strong rebound of the manufacturing and construction sectors; (ii) sizable fiscal support, followed by a more ambitious medium-term fiscal consolidation; and (iii) lower yet sizable current account deficit in 2020, followed by an improvement in 2021 and thereafter. Compared to the July 2021 DSA update during the fourth PCI review, henceforth the "July DSA update" the current projection takes into account the general SDR allocation and larger-than-expected Eurobond issuance in August 2021, which facilitated debt management. The main assumptions and projections are summarized in Text Table 3.

4. With the pandemic's impact on Rwanda's economy larger than estimated in 2020 and the recovery underway in 2021, the rebound is expected to be stronger than envisaged in the December 2020 DSA. After a 3.4 percent contraction in 2020, output is projected to grow by 10.2 percent in 2021. The projected rebound reflects base effects, stronger-than-anticipated performance in 2021H1 in industry and services; strong performance in 2021Q3 high frequency indicators for mining and manufacturing; the temporary and limited impact of mobility restrictions reinstituted in July; the acceleration in the vaccination rollout since July; recovery in external demand; and accommodative macroeconomic policies. Growth is expected to become more broad-based as the services sector recovers and returns to its pre-pandemic trend by 2023, subsequently converging to growth of 6.5 percent over the long term. The long-term growth (6.8 percent on average between 2026-41) under the baseline remains within the 10-year historical average of 7.1 percent once the large pandemic contraction in 2020 is excluded and is within the range of 6.5–7 percent

of where previous analyses have estimated Rwanda's potential GDP growth to be. Long-term growth in the baseline projections will continue to be driven by strong physical capital accumulation, but with investment shifting from public to private, reflecting NST policies to increase private sector participation, and followed by a gradual increase in human capital accumulation consistent with improvements in schooling and population growth. The negative growth in TFP is expected to be reversed and turn positive at around 1 percent. Given the strong deployment of support and policies to address potential scars from the pandemic and informed by recent analytical work, the growth impact of the pandemic scars is expected to be temporary with limited impact on long-term growth expected at this stage.[3]

5. The government's comprehensive package of policies to mitigate pandemic scars is expected to support growth in the medium to long term. The authorities expanded social safety nets to cushion households under the Economic Recovery Plan (ERP), reaching over 700 thousand households by September/October 2021. Social policies under the ERP are aimed at reversing human capital losses during the pandemic, notably by expanding programs to support poor pregnant and lactating women and reduce child stunting and malnutrition,; improving the quality and retention of primary and secondary schooling through the fast-tracking of the construction of new and less remote schools; expanding teacher training; and the adoption of digital technologies The authorities also deployed financial support to hard-hit firms under a revamped ERF (ERF-2), including through the new investment window that aims to support manufacturing and other strategic sectors to accelerate the recovery by creating jobs and improving Rwanda's external position. ERF-2 also seeks to reduce gender inequities exacerbated by the pandemic that constrain long-term growth by providing training and facilitating access to finance to female entrepreneurs.

6. The August 2021 Eurobond issuance will facilitate debt management and reduce liquidity risks in near term, while the general SDR allocation and a portion of the Eurobond proceeds are expected to finance a further increase in the fiscal deficit in FY2021/22. Part of the recently issued Eurobond proceeds amounting US$620 million was used to buy-back 85 percent of the Eurobond maturing in 2023 (US$374.5 million including associated fees and premium), alleviating liquidity risks, and to retire relatively expensive commercial loans (US$98 million), improving debt sustainability. The remaining Eurobond proceeds, amounting US$148 million, will be used to finance investment projects, which will add to the debt level. In addition, the authorities are planning to use a significant portion of the recent SDR allocation (about US$219 million) for budget financing. While the allocation per se does not affect debt sustainability, the baseline assumes that the authorities will spend 70 percent of their SDR resources (1.3 percent of GDP) in FY 2021/22 on a one-off basis for spending needs to respond to damage caused by a volcanic eruption near the border of the Democratic Republic of the Congo, to support social protection programs and social sector projects, which will add to the debt level.

7. Notwithstanding a larger fiscal deficit than envisaged at the December 2020 DSA to respond to the health, social, and economic fallout of the pandemic, debt is expected to gradually converge back to the anchor helped by the strong rebound in 2021 and a strong but growth-friendly fiscal consolidation. The COVID-19 induced fiscal support will contribute to higher near-term levels of public and publicly guaranteed (PPG) debt; however, the post-pandemic fiscal consolidation path, and the stronger than envisaged recovery underway in the first half of 2021 are projected to bring the total PPG debt back to the

[3] See Staff Report for 2021 Article IV Consultation and 5th Review under the Policy Coordination Instrument, Annex I.

Text Table 3. Key Macroeconomic and Debt Assumptions—Comparison with Previous Debt Sustainability Analyses[1]

Calendar year	2020	2021	2026	2031	2036	2041	2026-41
			Projections				
Selected indicators from the macro-framework and debt data							
(Percent, unless otherwise indicated)							
			PV of PPG External Debt to GDP Ratio				
2020 DSA (3rd PCI Review)	34.1	35.2	38.9	36.6	30.9	...	33.7
2021 DSA (4th PCI Review)	33.9	38.4	42.8	36.6	31.9	25.3	34.4
2021 DSA (5th PCI Review)	33.6	36.1	40.4	31.9	27.4	22.1	30.4
			PV of Public Debt to GDP Ratio				
2020 DSA (3rd PCI Review)	45.5	48.8	47.7	44.8	47.1	...	46.4
2021 DSA (4th PCI Review)	51.0	55.8	52.1	45.0	45.0	46.2	46.1
2021 DSA (5th PCI Review)	50.3	52.9	50.0	43.0	42.4	42.0	43.6
			Grant Element of New External Borrowing				
2020 DSA (3rd PCI Review)	48.3	47.5	44.3	40.3	31.4	...	35.5
2021 DSA (4th PCI Review)	0.0	29.8	38.0	31.5	30.0	18.6	32.5
2021 DSA (5th PCI Review)	-	26.3	39.2	31.0	31.0	19.3	33.4
			Stock of New Commercial Loan (billions of U.S. dollars)				
2020 DSA (3rd PCI Review)	0.0	0.0	0.9	1.0	1.1	...	1.1
2021 DSA (4th PCI Review)	0.0	0.5	0.9	0.6	1.4	2.6	1.3
2021 DSA (5th PCI Review)	0.0	0.6	0.8	0.4	1.3	2.6	1.1
			Real GDP Growth (annual percent change)				
2020 DSA (3rd PCI Review)	-0.2	5.7	6.1	7.2	6.8	...	6.8
2021 DSA (4th PCI Review)	-3.4	5.1	6.1	7.2	6.8	6.5	6.8
2021 DSA (5th PCI Review)	-3.4	10.2	6.1	7.2	6.8	6.5	6.8
			Current Account Balance (percent of GDP)				
2020 DSA (3rd PCI Review)	-12.2	-12.5	-7.6	-8.0	-7.5	...	-7.5
2021 DSA (4th PCI Review)	-12.2	-13.4	-6.8	-8.0	-7.5	-6.5	-7.4
2021 DSA (5th PCI Review)	-11.9	-11.0	-5.7	-7.5	-6.2	-5.1	-6.5
			Exports of goods and services (percent of GDP)				
2020 DSA (3rd PCI Review)	18.2	22.7	28.5	30.3	32.2	...	31.2
2021 DSA (4th PCI Review)	18.7	22.3	30.8	29.6	28.3	27.5	29.0
2021 DSA (5th PCI Review)	18.7	18.9	33.3	33.5	30.9	29.7	32.0
			Fiscal balance[2] (percent of GDP)				
2020 DSA (3rd PCI Review)	-9.7	-7.9	-3.7	-5.2	-4.8	...	-4.8
2021 DSA (4th PCI Review)	-9.4	-8.9	-3.2	-4.0	-3.3	-3.2	-3.5
2021 DSA (5th PCI Review)	-10.3	-9.3	-3.5	-4.1	-3.4	-3.3	-3.6

1/ Previous full DSA completed in January 2021; Update publish in July 2021.
Sources: Rwandan authorities; IMF and World Bank staff estimates and projections.

debt anchor of 65 percent of GDP by 2028, earlier than projected in the July DSA update.[4] Given the large FY 2021/22 fiscal deficit, this requires a large cumulative reduction in the overall headline deficit amounting 6.1 percentage points of GDP between FY 2022/23 and FY 2025/26 driven by the phase-off of COVID-19 and SDR-financed spending, tax policy and administration measures under the government medium-term revenue strategy spanning FY 2021/22 to FY 2023/24 (MTRS), and spending savings based on efficiency gains, digitalization in the delivery of public goods and services, and greater rationalization of public investment projects guided by the on-going World Bank's Public Expenditure Review (PER) and an IMF Public

[4] For discussions on the fiscal anchor please see the Third Review under the PCI staff report (Country Report No. 2021/001).

Investment Management Assessment (PIMA) planned for early in 2022. Domestic revenue mobilization measures have started to be legislated as part of the MTRS and are expected to be approved by Cabinet by end-2021. Spending measures will start to be adopted in the FY 2022/23 budget.

8. The DSA assumes continued support from bilateral and multilateral development partners over the medium term. The current fiscal framework foresees continued support to the NST implementation, while maintaining macroeconomic stability. In the near-term, financing needs of the government are expected to be met by increased support from official bilateral and multilateral partners, preserving concessionality of a significant portion of the debt. As of December 7, 2021, the World Bank disbursed US$98.2 million in FY2021/22 in the form of concessional financing, including budget support and project financing. out of US$991.2 million IDA allocation over this and the next fiscal year. From 2028 onwards, the financing mix is assumed to: (i) shift gradually away from concessional financing to market-based financing as Rwanda's income level rises, and (ii) shift from external to domestic financing as the local bond markets develop. The grant component of new external financing is assumed to decline as Rwanda develops. As a result, grant-equivalent new net external financing is projected to decline from 66 percent in 2022 and 52 percent in 2031 to under 35percent by 2041.[5]

9. Past public debt dynamics have been strongly affected by the materialization of fiscal risks and shocks to output (Figure 3). Changes in total public debt over the past five years have been driven by higher-than-anticipated primary deficits, unanticipated developments of the debt outside the budgetary central government, and lower-than-expected GDP growth—all leading to a higher-than-expected debt accumulation of almost 25 percentage points of GDP—well in excess of the median, and the upper edge of the 25–75 percent interquartile range, of other low-income countries. Hence the management of risks to mitigate future unanticipated fiscal developments outside the central government, which is a key pillar of the PCI, would help mitigate the prospects for unexpected debt increases. Going forward, the evolution of public debt will be dominated by the path of the primary fiscal deficit and real GDP growth.

10. Realism tools indicate that the planned fiscal adjustment is ambitious (Figure 4). The loosening of the fiscal deficit during FY2021/22 has rendered the required adjustment over the next few years more demanding during the post-pandemic period. A 3-year fiscal consolidation in the primary balance is expected to reach more than 5 percentage points of GDP from 2022 to 2025. Such adjustment lies well into the top quartile of the distribution of past adjustments for a sample of LICs, signaling that the envisaged fiscal adjustment in the baseline scenario is ambitious based on past experiences in LICs. While a significant share of the adjustment relates to phasing-out of one-off COVID-19-related spending, the credibility of fiscal adjustment of this magnitude hinges on the authorities' commitment to implement reforms and policies under the PCI program on domestic revenue mobilization and improvements in spending efficiency.

11. Rwanda's debt-carrying capacity continues to be assessed as "strong" (Text Tables 4a and 4b). The composite index (CI) for Rwanda, which measures the debt-carrying capacity in the new LIC-DSF, stands at 3.19, above the cut-off value of 3.05 for strong capacity countries. The underlying inputs for the

[5] This includes grants provided directly to the government as well as the grant element of new borrowing (difference between the face value and the PV of new debt).

calculation of the CI were sourced from the IMF's October 2021 WEO, and an update of the World Bank Country Policy and Institutional Assessment (CPIA) to 2020 levels. The CI score is largely driven by Rwanda's high CPIA score and adequate reserve coverage.[6]

Text Table 4a. Rwanda: Debt-Carrying Capacity

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.39	4.08	1.57	49%
Real growth rate (in percent)	2.72	6.49	0.18	6%
Import coverage of reserves (in percent)	4.05	38.01	1.54	48%
Import coverage of reserves^2 (in percent)	-3.99	14.45	-0.58	-18%
Remittances (in percent)	2.02	2.53	0.05	2%
World economic growth (in percent)	13.52	3.15	0.43	13%
CI Score			3.19	100%
CI rating			Strong	

Text Table 4b. Rwanda: Applicable Thresholds and Benchmarks

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23

TOTAL public debt benchmark	Weak	Medium	Strong
PV of total public debt in percent of GDP	35	55	70

DEBT SUSTAINABILITY ANALYSIS

External Debt

12. The external debt indicators suggest continued moderate risk of debt distress, but the solvency risks are rising, which makes the country more susceptible to external shocks compared to the December 2020 DSA (Tables 1 and 3; Figures 1 and 3).[7] Under the baseline scenario, the solvency indicators remain below their indicative thresholds. However, there are multiple breaches of the threshold under the most extreme shocks to exports, notably for the PV of external debt-to-exports ratio. This is mitigated by adequate international reserves and available external official financing. Similarly, the authorities' efforts to address climate related shocks, including through the 2019 National Environment and Climate Change Policy, Nationally Determined Contribution (NDC) goals, and the Green Growth and Climate Resilience Strategy, as well as their current efforts to mitigate pandemic scars to human capital are expected to mitigate shocks to growth going forward.

[6] This is based on the IMF's assessment of reserve adequacy.

[7] The LIC-DSF assesses the risk of debt distress by observing the evolution of selected indicators against predetermined thresholds that are set according to countries' debt-carrying capacities.

13. Liquidity indicators show that liquidity risks are muted, although significant gross financing needs make the country susceptible to adverse market conditions (Figures 1 and 6). The debt management strategy to smooth out the debt servicing profile helped to mitigate the liquidity risks, as the debt service-to-exports and debt service-to-revenue ratios, except for one-off breaches in 2021, remain below their thresholds throughout most of the period under all scenarios.[8] The breach in the external debt service indicator in 2021 reflects the Eurobond buyback operation, which took place in August this year and helped to smooth the debt service for the Eurobond bullet payments due in 2023; the country, however, will need to address the Eurobond rollover risks in 2031. At the same time, Rwanda may still face liquidity pressures due to adverse market conditions, as pointed out by a breach of the market-financing risk indicator for gross financing needs (Figure 6). Similar to the solvency indicators, these risks are expected to be partly mitigated by the adequate international reserves.

14. The PV of external debt-to-GDP ratio increases sharply under the historical scenario since the latter assumes the recurrence of several large external shocks and large external imbalances observed in the past (Table 3 and Figure 1). PV of debt-to-GDP ratio rises sharply under the historical scenario. This is primarily due to the large current account deficit and negative USD GDP deflator calibrated using historical averages, which covered a period including several large shocks (donor withdrawal, commodity prices, and drought) as well as large external imbalances, which were corrected over 2015–17, primarily through a large exchange rate adjustment, as envisaged under the 2016 PSI/SCF-supported program. The large current account deficit and the negative GDP deflator account for almost all of the divergence between the baseline and historical scenarios for the solvency indicators. Policy adjustment to ensure a steady narrowing of the current account deficit from its elevated 2020–21 levels, as envisaged under the baseline scenario (Text Table 2) is key to strengthen robustness of the debt dynamics.

Public Debt

15. Total PPG debt dynamics indicate that solvency risks have increased relative to the 2020 DSA and that buffers to cushion the impact of major shocks remain limited (Tables 2 and 4; Figures 2 and 3). Faster debt accumulation has resulted in less favorable near-term debt dynamics relative to the 2020 DSA, even as risks remain moderate. In the baseline scenario, the PV of PPG debt remains below the indicative benchmark of 70 percent: it is projected to peak at 56.1 percent of GDP in 2023 before reaching the East African Community Monetary Union Protocol for debt convergence criterion of PV of 50 percent in 2026. In addition, the PV of PPG debt does not cross the 70 percent threshold even under severe shocks. The nominal PPG debt-to-GDP ratio (including the use of the recent SDR allocation for budget financing) is projected to peak at 77.9 percent in 2023 and reach the nominal debt anchor of 65 percent in 2028 in the baseline scenario, conditional on successful implementation of policies to support the ambitious fiscal consolidation. On the other hand, similar to the external debt analysis, while liquidity risks are muted in the baseline, in part reflecting a smoother debt service path achieved by this year's Eurobond issuance, debt service costs would remain elevated under severe growth and other country-specific shocks and may also be sensitive to adverse market conditions.

[8] According to the LIC DSF guidance note, single short-lived breaches (1-year) are assumed not to affect the risk rating.

ASSESSMENT

16. Rwanda's debt is assessed to be sustainable with a moderate risk of external and overall public debt distress.[9] Furthermore, a granular assessment of the moderate risk rating shows that Rwanda has limited space to absorb shocks (Figure 5). Relative to the last full DSA in 2020, the risk assessment remained unchanged; however, solvency indicators deteriorated somewhat due to larger debt accumulation in the near term, while liquidity risks have been reduced. Given the moderate risk of debt distress assessment, the limit on the stock of new external PPG debt will continue to be monitored under the PCI.

17. However, updates with respect to the economic impact and policy response to the COVID-19 crisis are evolving and risks to the debt outlook and sustainability could deteriorate should the pandemic shock prolong further. The current macroeconomic framework which underpins this DSA reflects currently available information. The baseline scenario assumes Rwanda gradually reverts to its pre-pandemic growth trend and continues to achieve robust growth over the medium term, while concessional financing is expected to decline gradually in the long term. Main risks to this outlook are external shocks to growth and/or exports, and worse-than-expected external financing conditions, which can be aggravated by a prolonged pandemic and uneven recovery.

18. Rwanda's debt challenges require robust and credible plans to ensure long term fiscal sustainability without jeopardizing short term recovery efforts. Fiscal reforms proposed for the implementation of the World Bank's Sustainable Development Finance Policy (SDFP) in FY 2021/22 complement the government's fiscal consolidation efforts under the IMF PCI program, and are designed to follow a sequenced intervention logic: a near term focus on strengthening the foundations of fiscal risk management and transparency followed by a growth-friendly fiscal consolidation once the crisis abates. In particular,

- **Fiscal risk management and transparency.** Improved budgetary control is essential given Rwanda's past record of significant debt surprises, which made it an outlier compared to low-income country peers. Strengthening the identification, assessment, and management of fiscal risks, one of the pillars under the PCI, remains central to mitigating debt surprises. With technical assistance and support from IMF and the World Bank, the authorities have shown progress in these areas and have indicated their commitment to the required reforms going forward, including those to publish fiscal risk statements and strengthen the fiscal transparency, oversight, and management of SOEs and PPPs.[10] Sustaining on-going efforts to improving the functioning of the domestic bond market will also be important to address worse-than-expected external financing conditions.[11]

[9] This assessment is in line with the streamlined DSA update under the fourth PCI review in 2021 (the July DSA update) (July 2021, IMF Country Report 21/164).

[10] With IMF TA support, a new Organic Budget Law expected to be approved in 2022Q1 will mandate the annual publication of fiscal risk statements. With World Bank support, authorities have committed to publish enhanced semi-annual debt statistics bulletins with expanded coverage of non-guaranteed outstanding SOE debt.

[11] Rwanda has a 10-year Capital Market Master Plan aimed at developing capital markets and benefits from ongoing TA from the World Bank and IFC through the capital market development program which aims, inter alia, at improving the framework for government securities market, strengthening liquidity in the secondary market, and developing the investor base through capacity building and support to regulators.

- **Fiscal consolidation.** Government's plans to adopt a credible fiscal consolidation as soon as the COVID-19 crisis abates are key to facilitating a steady decline in the debt ratio and restore fiscal buffers. This is even more important as realism tests raise stronger concerns (compared to the previous full DSA) about the credibility of the fiscal adjustments in the baseline, which must now be achieved over a shorter horizon as the budget deficit for the current fiscal year has been loosened. In this context, it would be important to ensure that COVID-support measures will be strictly one-off, and adequate structural revenue and expenditure measures should be promptly implemented to safeguard the credibility of the adjustment in reversing recent debt dynamics and addressing external imbalances. On the revenue side, this will require the timely implementation of the MTRS and the unwinding of temporary tax exemptions to support the post-pandemic recovery. On the spending side, measures should aim at rationalizing spending through efficiency gains, digitalization, and better prioritization of the existing pipeline of public projects through the adoption of a medium-term investment program.

Authorities' Views

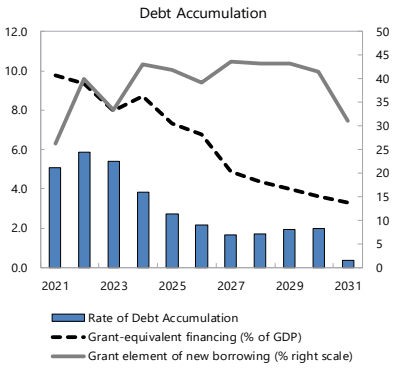
19. The authorities broadly agree with the results of the DSA and the overall assessment of a moderate risk of external and overall debt distress. They expect their current liability management strategy of converting short term external debt into long term domestic debt to shield the portfolio from refinancing risk stemming from external shocks. They also consider the recent Eurobond buyback of 85 percent of the existing US$400 million Eurobond with bullet payment in 2023 at favorable terms to significantly reduce the liquidity risks in the near term. Going forward, the authorities' debt management strategy will continue to be based on maximizing external concessional funding to avoid pressures on the debt repayment profile, while also implementing prudent spending and revenue measures agreed under the PCI to bring the debt level to the anchor of 65 percent debt-to-GDP, which they expect to build buffers to absorb shocks and reduce solvency risks.

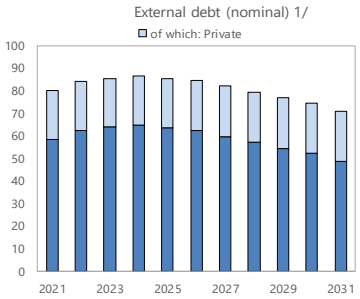
Table 1. Rwanda: External Debt Sustainability Framework, Baseline Scenario, 2020–41

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**76.6**	**79.9**	**84.1**	**85.6**	**86.6**	**85.5**	**84.7**	**70.8**	**52.5**	**42.6**	**80.9**
of which: public and publicly guaranteed (PPG)	**55.6**	**58.6**	**62.4**	**63.9**	**64.7**	**63.5**	**62.2**	**48.7**	**30.7**	**31.4**	**58.9**
Change in external debt	13.6	3.3	4.1	1.5	1.0	-1.0	-0.9	-3.7	-1.7		
Identified net debt-creating flows	**11.1**	**1.5**	**2.9**	**0.7**	**-0.6**	**-1.6**	**-1.9**	**-0.9**	**-3.5**	**6.5**	**-0.3**
Non-interest current account deficit	**9.8**	**8.7**	**9.1**	**7.9**	**6.4**	**5.4**	**3.1**	**5.2**	**2.9**	**9.5**	**5.9**
Deficit in balance of goods and services	16.0	16.7	16.5	14.5	14.0	12.5	10.2	8.6	2.9	**15.7**	**11.9**
Exports	18.7	18.9	24.4	26.7	27.7	31.5	33.3	33.5	29.7		
Imports	34.6	35.6	40.9	41.2	41.7	44.0	43.5	42.1	32.6		
Net current transfers (negative = inflow)	-6.0	-8.0	-7.3	-6.5	-7.7	-7.2	-7.1	-4.1	-1.5	**-7.4**	**-6.2**
of which: official	-5.8	-7.4	-7.2	-6.3	-7.0	-6.2	-6.1	-2.6	-0.9		
Other current account flows (negative = net inflow)	-0.2	-0.1	0.0	0.0	0.1	0.2	0.1	0.6	1.6	**1.2**	**0.2**
Net FDI (negative = inflow)	**-1.0**	**-2.2**	**-3.2**	**-3.5**	**-3.6**	**-3.6**	**-2.7**	**-3.5**	**-5.4**	**-2.6**	**-3.2**
Endogenous debt dynamics 2/	**2.3**	**-5.0**	**-3.0**	**-3.7**	**-3.4**	**-3.4**	**-2.3**	**-2.6**	**-1.0**		
Contribution from nominal interest rate	2.2	2.3	2.3	2.4	2.6	2.6	2.6	2.3	2.2		
Contribution from real GDP growth	2.1	-7.3	-5.4	-6.1	-5.9	-6.0	-4.9	-4.9	-3.2		
Contribution from price and exchange rate changes	-2.0	...	...	...	...	...	...	...	...		
Residual 3/	**2.5**	**1.8**	**1.2**	**0.7**	**1.6**	**0.6**	**1.0**	**-2.8**	**1.8**	**-0.5**	**-0.2**
of which: exceptional financing	-0.2	-0.3	-0.1	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**33.6**	**36.1**	**39.2**	**40.9**	**41.5**	**40.9**	**40.4**	**31.9**	**22.1**		
PV of PPG external debt-to-exports ratio	**179.7**	**190.9**	**160.5**	**153.3**	**149.9**	**130.0**	**121.1**	**95.1**	**74.5**		
PPG debt service-to-exports ratio	**10.6**	**34.5**	**7.8**	**9.0**	**8.2**	**8.5**	**8.3**	**12.8**	**9.6**		
PPG debt service-to-revenue ratio	**10.6**	**34.4**	**9.9**	**12.9**	**11.9**	**13.6**	**14.1**	**19.4**	**11.5**		
Gross external financing need (Billion of U.S. dollars)	1.5	1.9	1.4	1.4	1.3	1.3	1.2	2.5	2.6		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-3.4	10.2	7.2	7.9	7.5	7.5	6.1	7.2	6.5	**6.1**	**7.4**
GDP deflator in US dollar terms (change in percent)	3.2	-2.9	-0.2	1.0	0.3	0.5	0.7	2.0	2.0	**-0.6**	**0.8**
Effective interest rate (percent) 4/	3.4	3.2	3.1	3.1	3.2	3.2	3.2	3.4	4.5	**3.5**	**3.2**
Growth of exports of G&S (US dollar terms, in percent)	-14.4	8.4	38.2	19.2	11.8	22.8	13.1	6.3	8.1	**11.4**	**14.5**
Growth of imports of G&S (US dollar terms, in percent)	-4.3	10.2	22.9	9.6	9.3	13.9	5.7	5.2	6.3	**8.5**	**10.3**
Grant element of new public sector borrowing (in percent)	...	26.3	39.8	33.4	43.0	41.8	39.2	31.0	19.3	**...**	**38.7**
Government revenues (excluding grants, in percent of GDP)	18.8	19.0	19.1	18.7	19.1	19.7	19.7	22.3	25.0	**14.9**	**20.4**
Aid flows (in Billion of US dollars) 5/	0.5	1.2	1.3	1.1	1.4	1.2	1.2	0.9	0.9		
Grant-equivalent financing (in percent of GDP) 6/	...	9.8	9.3	8.0	8.7	7.3	6.8	3.3	1.3	**...**	**6.4**
Grant-equivalent financing (in percent of external financing) 6/	...	50.9	65.9	59.1	69.4	69.6	70.3	52.1	34.5	**...**	**63.7**
Nominal GDP (Billion of US dollars)	10	11	12	13	14	15	16	25	58		
Nominal dollar GDP growth	-0.2	7.0	7.0	8.9	7.8	8.1	6.8	9.3	8.6	**5.4**	**8.3**
Memorandum items:											
PV of external debt 7/	54.6	57.5	60.9	62.6	63.4	63.0	62.8	54.0	44.0		
In percent of exports	292.4	303.9	249.4	234.4	228.8	199.9	188.3	160.9	148.1		
Total external debt service-to-exports ratio	31.2	55.9	24.5	24.3	23.1	21.7	21.0	25.3	23.5		
PV of PPG external debt (in Billion of US dollars)	3.5	4.0	4.6	5.3	5.8	6.1	6.5	7.9	12.8		
(PVt-PVt-1)/GDPt-1 (in percent)		5.1	5.9	5.4	3.8	2.7	2.2	0.4	0.9		
Non-interest current account deficit that stabilizes debt ratio	-3.8	5.4	5.0	6.5	5.4	6.4	4.0	8.9	4.6		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

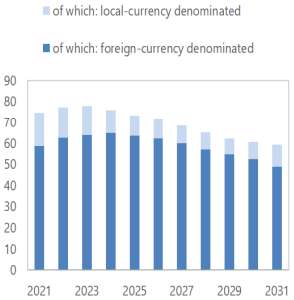
Table 2. Rwanda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020–41

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**71.3**	**74.6**	**77.2**	**77.9**	**75.7**	**73.1**	**71.1**	**59.1**	**50.1**	**40.7**	**69.5**
of which: external debt	**55.6**	**58.6**	**62.4**	**63.9**	**64.7**	**63.5**	**62.2**	**48.7**	**30.7**	**31.4**	**58.9**
Change in public sector debt	14.6	3.3	2.6	0.7	-2.2	-2.6	-2.0	-1.4	-0.6		
Identified debt-creating flows	**10.0**	**3.5**	**2.8**	**0.9**	**-2.0**	**-2.3**	**-1.7**	**-1.3**	**-0.6**	**2.0**	**-0.9**
Primary deficit	**8.7**	**7.3**	**5.5**	**6.0**	**2.2**	**2.0**	**1.7**	**2.6**	**1.7**	**3.1**	**3.1**
Revenue and grants	23.6	25.4	25.3	23.9	24.9	24.8	24.7	24.2	25.7	**21.3**	**24.6**
of which: grants	4.8	6.4	6.1	5.2	5.8	5.0	4.9	1.9	0.7		
Primary (noninterest) expenditure	32.2	32.7	30.8	29.9	27.1	26.7	26.3	26.8	27.4	**24.4**	**27.7**
Automatic debt dynamics	**1.4**	**-3.8**	**-2.8**	**-5.1**	**-4.1**	**-4.2**	**-3.2**	**-3.9**	**-2.3**		
Contribution from interest rate/growth differential	1.7	-6.0	-4.4	-5.1	-5.1	-4.8	-3.8	-3.9	-2.3		
of which: contribution from average real interest rate	-0.2	0.6	0.6	0.6	0.3	0.5	0.4	0.1	0.8		
of which: contribution from real GDP growth	2.0	-6.6	-5.0	-5.6	-5.4	-5.3	-4.2	-4.0	-3.1		
Contribution from real exchange rate depreciation	-0.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**-0.2**	**-0.1**	**-0.1**	**0.0**	**0.0**	**-0.1**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (use of earmarked fund)	0.0	0.0	0.0	0.0	-0.2	-0.1	-0.1	0.0	0.0		
Residual	**4.5**	**2.0**	**1.4**	**-0.3**	**0.8**	**0.3**	**0.2**	**0.0**	**0.0**	**3.2**	**0.3**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**50.3**	**52.9**	**55.3**	**56.1**	**53.5**	**51.4**	**50.0**	**43.0**	**42.0**		
PV of public debt-to-revenue and grants ratio	**213.8**	**208.4**	**218.6**	**234.8**	**215.0**	**207.5**	**203.0**	**177.6**	**163.5**		
Debt service-to-revenue and grants ratio 3/	**40.8**	**55.7**	**35.7**	**39.1**	**36.3**	**31.6**	**31.2**	**37.7**	**38.7**		
Gross financing need 4/	18.3	21.4	14.6	15.3	11.0	9.6	9.2	11.7	11.7		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-3.4	10.2	7.2	7.9	7.5	7.5	6.1	7.2	6.5	**6.1**	**7.4**
Average nominal interest rate on external debt (in percent)	1.5	1.4	1.4	1.5	1.6	1.6	1.6	1.5	2.1	**2.0**	**1.5**
Average real interest rate on domestic debt (in percent)	-0.7	7.5	5.5	3.8	5.2	6.0	6.2	4.6	4.3	**1.3**	**5.5**
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.8	...	...	...	...	...	...	...	...	**2.4**	**...**
Inflation rate (GDP deflator, in percent)	8.3	2.3	5.3	7.1	5.5	5.3	5.0	5.0	5.0	**4.3**	**5.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	28.3	11.8	1.0	4.6	-2.5	5.9	4.6	7.2	7.3	**11.1**	**5.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-5.9	4.0	3.0	5.3	4.4	4.6	3.6	4.0	2.3	**-2.5**	**4.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

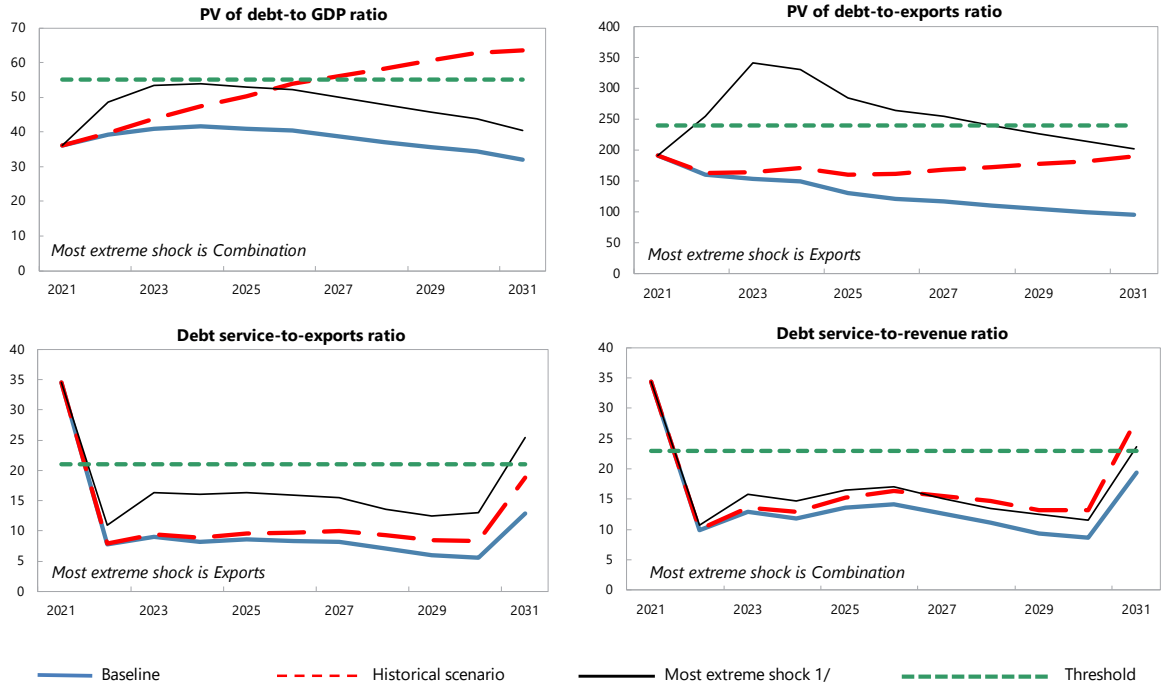
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Rwanda: Indicators of Public and Publicly-Guaranteed External Debt Under Alternative Scenarios, 2021–31 1/ 2/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.9%	1.9%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	6	6

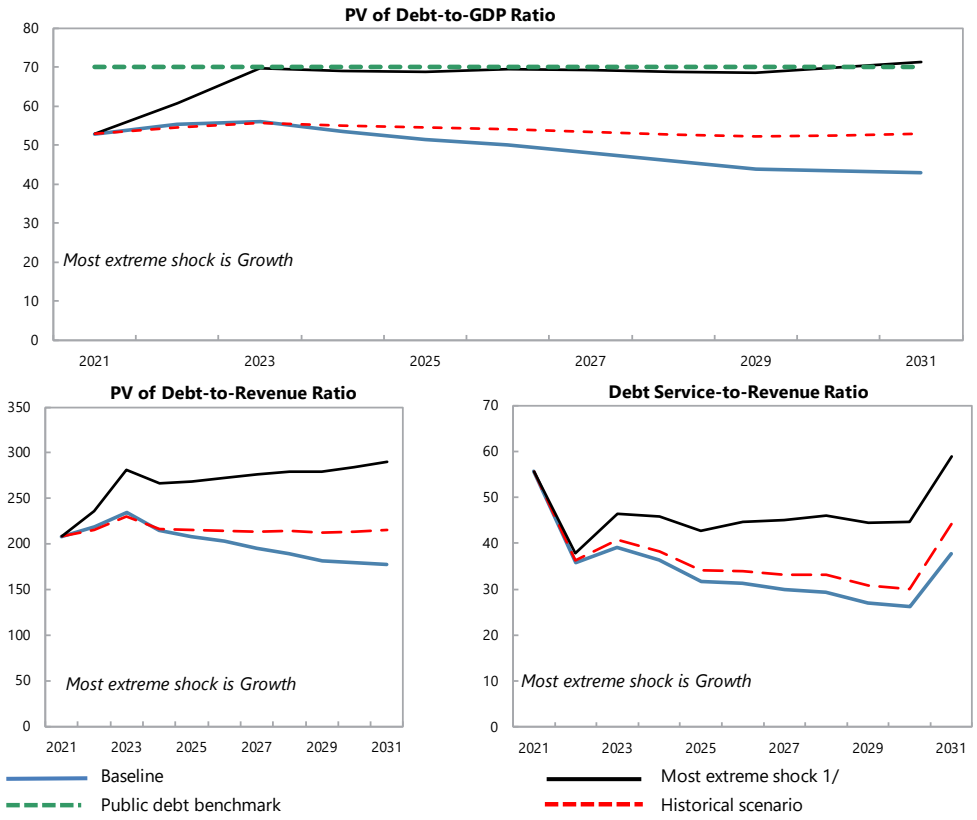
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Rwanda: Indicators of Public Debt Under Alternative Scenarios, 2021–31 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	52%	52%
Domestic medium and long-term	17%	17%
Domestic short-term	31%	31%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.9%	1.9%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	5.4%	5.4%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	1%	1.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2021–31

(In percent)

	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	**36.1**	**39.2**	**40.9**	**41.5**	**40.9**	**40.4**	**38.8**	**37.1**	**35.7**	**34.5**	**31.9**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	36.1	39.8	43.8	47.3	50.3	53.9	56.0	58.3	60.6	62.8	63.5
B. Bound Tests											
B1. Real GDP growth	36.1	42.3	48.2	48.9	48.2	47.5	45.7	43.7	42.0	40.6	37.5
B2. Primary balance	36.1	40.0	44.0	44.7	44.2	43.8	42.1	40.4	38.8	37.4	34.7
B3. Exports	36.1	44.1	53.2	53.4	52.4	51.5	49.5	47.3	45.2	43.1	39.6
B4. Other flows 2/	36.1	41.4	45.6	46.0	45.2	44.6	42.8	40.9	39.2	37.7	34.8
B6. One-time 30 percent nominal depreciation	36.1	49.6	47.7	48.6	48.0	47.4	45.5	43.5	41.9	40.7	37.7
B6. Combination of B1-B5	36.1	48.5	53.3	53.8	52.9	52.1	50.0	47.9	45.7	43.8	40.4
C. Tailored Tests											
C1. Combined contingent liabilities	36.1	41.6	44.0	44.7	44.4	44.0	42.4	40.6	39.1	37.8	35.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	36.1	43.8	45.8	46.6	45.7	44.8	43.1	41.3	39.7	38.4	35.5
Threshold	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of debt-to-exports ratio											
Baseline	**190.9**	**160.5**	**153.3**	**149.9**	**130.0**	**121.1**	**116.5**	**109.8**	**104.5**	**100.0**	**95.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	190.9	162.8	164.0	170.7	159.9	161.6	168.1	172.6	177.5	182.2	189.5
B. Bound Tests											
B1. Real GDP growth	190.9	160.5	153.3	149.9	130.0	121.1	116.5	109.8	104.5	100.0	95.1
B2. Primary balance	190.9	163.8	164.5	161.4	140.3	131.2	126.5	119.4	113.7	108.6	103.5
B3. Exports	190.9	254.4	341.2	330.0	284.8	264.6	254.5	239.8	226.9	213.9	202.4
B4. Other flows 2/	190.9	169.7	170.6	165.9	143.6	133.6	128.5	121.1	115.0	109.3	103.7
B6. One-time 30 percent nominal depreciation	190.9	160.5	141.1	138.5	120.3	112.2	107.9	101.8	97.0	93.2	88.9
B6. Combination of B1-B5	190.9	242.8	174.5	255.6	221.1	205.8	197.9	186.6	176.4	167.5	158.8
C. Tailored Tests											
C1. Combined contingent liabilities	190.9	170.3	164.6	161.4	141.0	131.9	127.3	120.3	114.6	109.7	104.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	190.9	160.5	153.4	150.4	129.8	120.3	115.7	109.4	104.2	99.7	94.8
Threshold	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**
Debt service-to-exports ratio											
Baseline	**34.5**	**7.8**	**9.0**	**8.2**	**8.5**	**8.3**	**8.1**	**7.1**	**6.0**	**5.5**	**12.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	34.5	7.9	9.5	8.8	9.6	9.7	10.0	9.3	8.4	8.4	18.7
B. Bound Tests											
B1. Real GDP growth	34.5	7.8	9.0	8.2	8.5	8.3	8.1	7.1	6.0	5.5	12.8
B2. Primary balance	34.5	7.8	9.3	8.7	9.0	8.8	8.6	7.5	6.4	6.0	13.4
B3. Exports	34.5	11.0	16.4	16.1	16.3	15.8	15.4	13.5	12.4	13.0	25.4
B4. Other flows 2/	34.5	7.8	9.2	8.6	8.9	8.7	8.5	7.4	6.5	6.3	13.6
B6. One-time 30 percent nominal depreciation	34.5	7.8	9.0	7.9	8.3	8.1	7.9	6.9	5.8	4.9	12.3
B6. Combination of B1-B5	34.5	10.3	14.5	13.3	13.6	13.2	12.9	11.3	10.5	9.7	20.7
C. Tailored Tests											
C1. Combined contingent liabilities	34.5	7.8	9.3	8.5	8.8	8.6	8.4	7.4	6.2	5.7	13.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	34.5	7.8	9.2	9.4	10.8	9.5	7.0	6.0	5.8	5.5	12.8
Threshold	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**
Debt service-to-revenue ratio											
Baseline	**34.4**	**9.9**	**12.9**	**11.9**	**13.6**	**14.1**	**12.6**	**11.2**	**9.3**	**8.6**	**19.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	34.4	10.1	13.5	12.8	15.3	16.4	15.5	14.6	13.2	13.1	28.3
B. Bound Tests											
B1. Real GDP growth	34.4	10.7	15.2	14.0	16.0	16.6	14.9	13.2	11.0	10.1	22.8
B2. Primary balance	34.4	9.9	13.3	12.7	14.3	14.8	13.3	11.8	10.1	9.5	20.2
B3. Exports	34.4	10.0	13.7	13.6	15.2	15.6	14.0	12.4	11.4	11.9	22.4
B4. Other flows 2/	34.4	9.9	13.2	12.5	14.2	14.6	13.1	11.6	10.2	9.8	20.5
B6. One-time 30 percent nominal depreciation	34.4	12.6	16.4	14.5	16.7	17.3	15.5	13.7	11.5	9.8	23.5
B6. Combination of B1-B5	34.4	10.7	15.8	14.6	16.5	17.0	15.2	13.4	12.5	11.5	23.7
C. Tailored Tests											
C1. Combined contingent liabilities	34.4	9.9	13.3	12.3	14.0	14.5	13.0	11.6	9.7	9.0	19.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	34.4	9.9	13.1	13.6	17.2	16.1	10.9	9.5	9.2	8.6	19.3
Threshold	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Includes official and private transfers and FDI.

Table 4. Rwanda: Sensitivity Analysis for Key Indicators of Public Debt, 2021–31

(In percent)

	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	**52.9**	**55.3**	**56.1**	**53.5**	**51.4**	**50.0**	**47.9**	**45.9**	**43.9**	**43.3**	**43.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	53	55	56	55	55	54	53	53	52	52	53
B. Bound Tests											
B1. Real GDP growth	53	61	70	69	69	69	69	69	69	70	71
B2. Primary balance	53	57	61	58	56	54	52	49	47	46	46
B3. Exports	53	60	68	65	63	61	59	56	53	52	51
B4. Other flows 2/	53	58	61	58	56	54	52	50	48	47	46
B6. One-time 30 percent nominal depreciation	53	61	59	55	51	49	45	42	39	38	36
B6. Combination of B1-B5	53	55	59	57	55	54	53	51	49	49	49
C. Tailored Tests											
C1. Combined contingent liabilities	53	61	61	58	56	54	52	50	47	47	46
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	53	55	56	54	51	50	48	46	44	43	43
Public debt benchmark	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**
PV of Debt-to-Revenue Ratio											
Baseline	**208.4**	**218.6**	**234.8**	**215.0**	**207.5**	**203.0**	**195.3**	**189.4**	**181.6**	**179.4**	**177.6**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	208	215	230	216	215	215	214	214	213	214	215
B. Bound Tests											
B1. Real GDP growth	208	236	281	266	268	272	276	279	279	285	290
B2. Primary balance	208	226	255	233	225	219	211	204	195	192	190
B3. Exports	208	238	286	263	254	248	239	231	221	215	209
B4. Other flows 2/	208	228	255	233	225	220	212	205	197	193	190
B6. One-time 30 percent nominal depreciation	208	245	252	225	211	201	187	176	164	157	150
B6. Combination of B1-B5	208	220	247	228	222	219	214	209	203	201	201
C. Tailored Tests											
C1. Combined contingent liabilities	208	241	256	234	225	220	212	205	196	193	191
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	208	219	235	216	207	202	194	189	181	179	177
Debt Service-to-Revenue Ratio											
Baseline	**55.7**	**35.7**	**39.1**	**36.3**	**31.6**	**31.2**	**29.9**	**29.4**	**27.1**	**26.2**	**37.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	56	36	41	38	34	34	33	33	31	30	44
B. Bound Tests											
B1. Real GDP growth	56	38	46	46	43	45	45	46	45	45	59
B2. Primary balance	56	36	42	41	35	35	32	31	29	28	39
B3. Exports	56	36	40	38	33	32	31	30	29	29	40
B4. Other flows 2/	56	36	39	37	32	32	30	30	28	27	39
B6. One-time 30 percent nominal depreciation	56	35	40	36	33	32	31	30	27	26	40
B6. Combination of B1-B5	56	35	41	40	35	35	34	34	31	30	42
C. Tailored Tests											
C1. Combined contingent liabilities	56	36	49	40	37	35	32	31	29	27	39
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	56	36	39	38	35	33	28	28	27	26	38

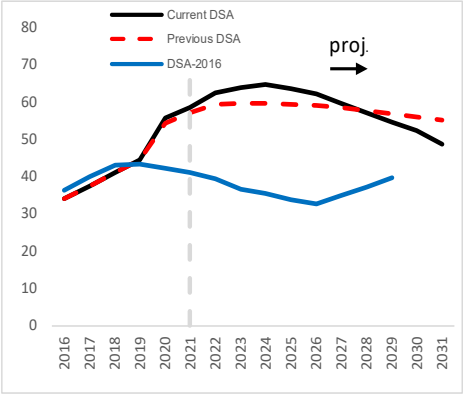
Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

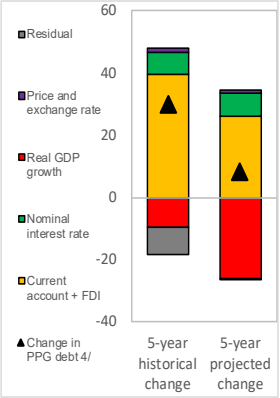
2/ Includes official and private transfers and FDI.

Figure 3. Rwanda: Drivers of Debt Dynamics—Baseline Scenario[1/]

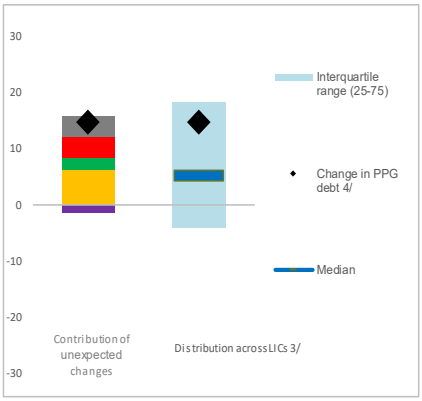
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

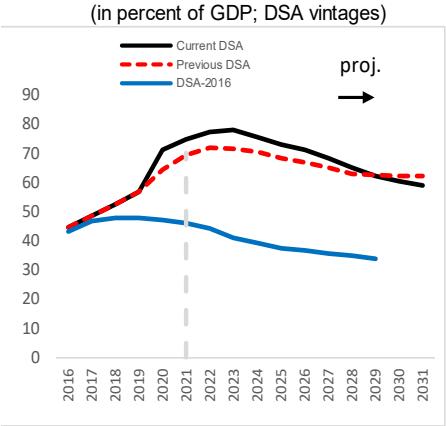


Unexpected Changes in Debt 2/
(past 5 years, percent of GDP)

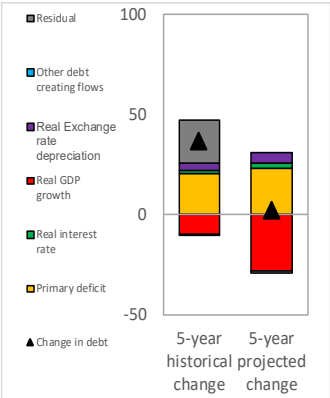


Public debt

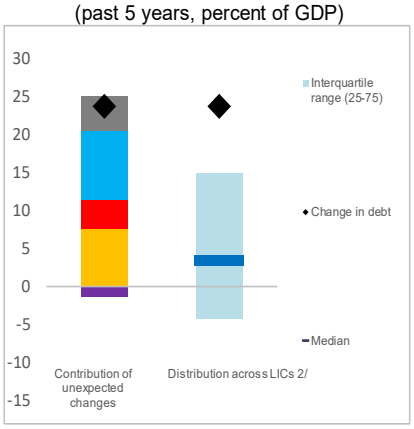
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Compared to 2016 DSA and the previous full DSA in 2020.

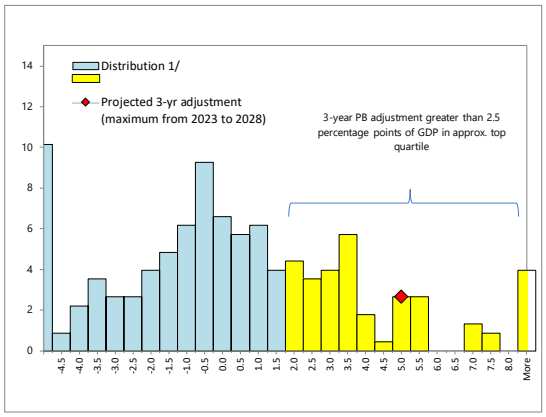
2/ Difference between anticipated and actual contributions on debt ratios.

3/ Distribution across LICs for which LIC DSAs were produced.

4/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

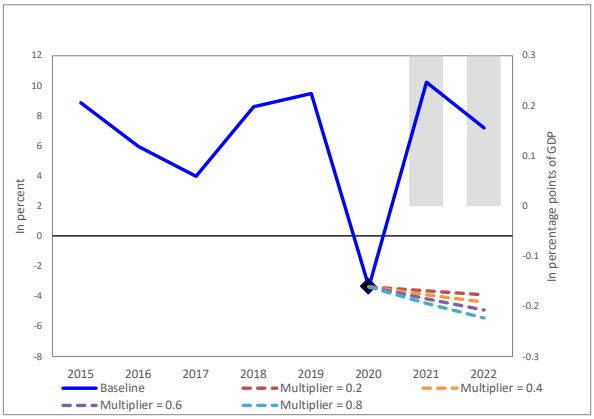
Figure 4. Rwanda: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



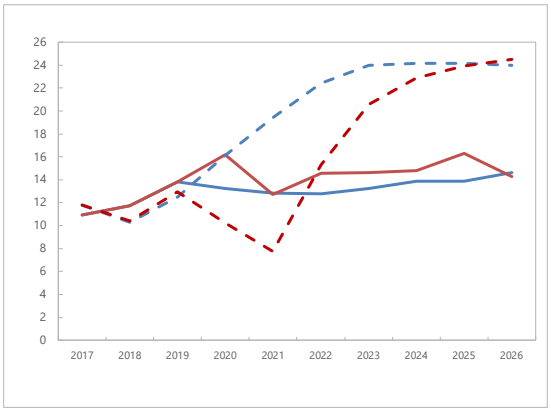
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

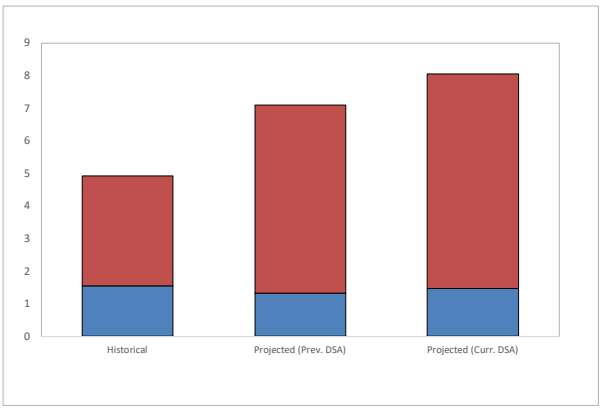
Public and Private Investment Rates 1/
(% of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA

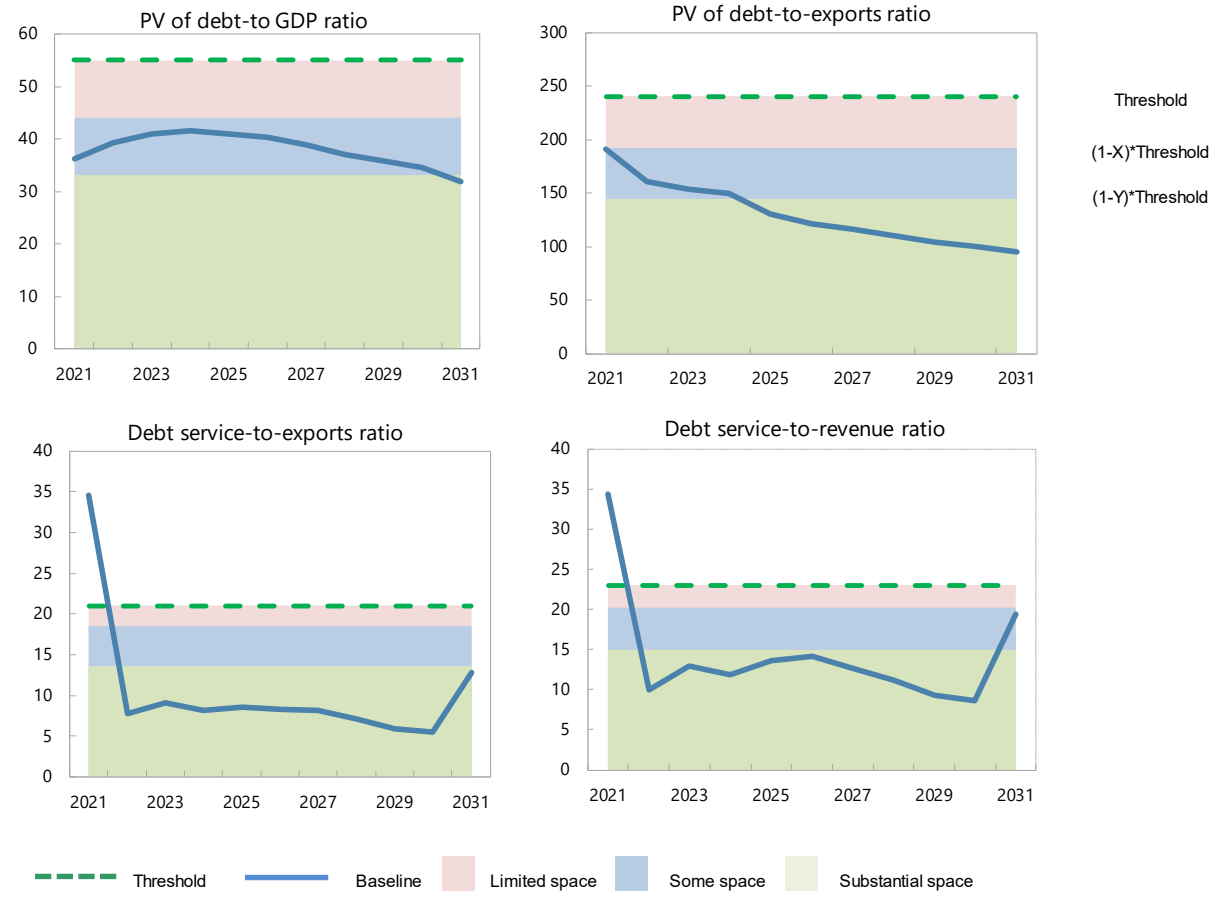
1/ Private investment includes the Bugesera airport project.

Contribution to Real GDP growth
(percent, 5-year average)



Contribution of other factors
Contribution of government capital

Figure 5. Rwanda: Qualification of the Moderate Category, 2020–31[1/]



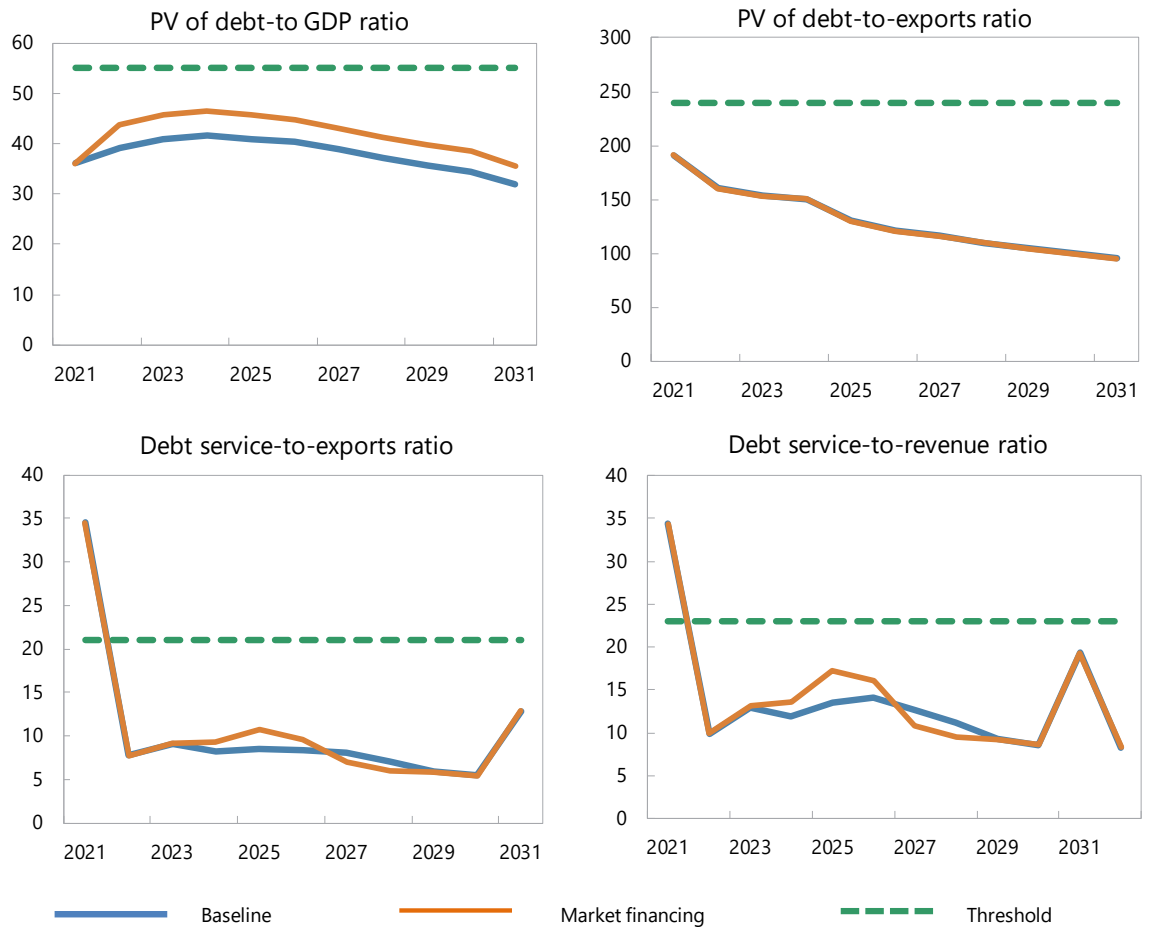
Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Figure 6. Rwanda: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	21	n.a.
Breach of benchmark	Yes	n.a.
Potential heightened liquidity needs	High	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.